MINOR PROJECT ASSESSMENT

Cybersecurity & Networking Fundamentals • Foundation Batch

SECTION A — CYBERSECURITY FUNDAMENTALS

15 MARKS

Q1. Fundamentals (5 Marks)

Cybersecurity is the practice of protecting systems, networks, devices, software applications, and critical data from unauthorized access, malicious attacks, theft, or damage. It is critical for safeguarding sensitive personal data (PII), protecting business continuity, avoiding financial losses/penalties, and maintaining public trust.

The CIA Triad

- **Confidentiality:** Restricting access to authorized users only.
  Example: Encrypting user databases using AES-256.
- **Integrity:** Ensuring data remains accurate and unaltered.
  Example: Using SHA-256 hashes to verify file downloads.
- **Availability:** Ensuring systems are online when needed.
  Example: Redundant servers and load balancing against DDoS.



Q2. Security Thinking (5 Marks)

Term	Definition	Practical Example
Vulnerability	A system flaw or software weakness that can be exploited.	Unpatched software vulnerability on an old web server.
Threat	Any actor or event with the potential to cause damage.	A cybercriminal group looking to deploy ransomware.
Risk	Potential loss when a threat exploits a vulnerability.	Financial & reputational damage from a data breach.
Attack	The intentional act of exploiting a vulnerability.	Executing an SQL Injection script to leak passwords.

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Q3. Types of Security (5 Marks)

1. Network Security: Secures network infrastructure and data traffic using firewalls, VPNs, and intrusion prevention systems.

2. Application Security: Focuses on fixing code vulnerabilities during software development via code reviews and input verification.

3. Cloud Security: Enforces access controls, policies, and encryption to secure cloud applications and data (e.g., AWS, Azure).

4. Endpoint Security: Protects end-user hardware like laptops and mobile devices via antivirus, EDR software, and full encryption.

5. Physical Security: Controls real-world physical access to facilities and data centers using security guards, biometric locks, and CCTV surveillance.

SECTION B — NETWORKING FUNDAMENTALS 20 MARKS

Q4. What is Networking? (5 Marks)

A **computer network** is an interconnected collection of autonomous devices capable of sharing hardware resources, files, and digital communication using standardized protocols. Organizations rely on networks for central data management, hardware cost savings, seamless team collaboration, and real-time remote communication.

Network Type	Geographic Coverage	Practical Example
PAN (Personal Area Network)	Very small personal space (<10 meters)	Bluetooth connection between a phone and wireless earbuds.
LAN (Local Area Network)	Single office, home, or school building	Wi-Fi/Ethernet network inside a high school computer lab.
MAN (Metropolitan Area Network)	Entire town, city, or multi-building campus	Municipal fiber-optic network linking municipal offices.
WAN (Wide Area Network)	Broad regional, national, or global scale	The global **Internet** or corporate links across countries.

Q5. Network Devices (5 Marks)

- **Hub:** Layer 1 device that connects host computers; blindly broadcasts received frames out to all physical ports.
- **Switch:** Layer 2 device that intelligently forwards local traffic based on destination MAC addresses.
- **Router:** Layer 3 device that interconnects different IP subnets and routes packets across networks using IP addresses.
- **Firewall:** Security gateway operating at Layers 3–7 that monitors and controls traffic using security rules.
- **Access Point (AP):** Layer 2 wireless transceiver allowing Wi-Fi clients to connect to a wired network infrastructure.

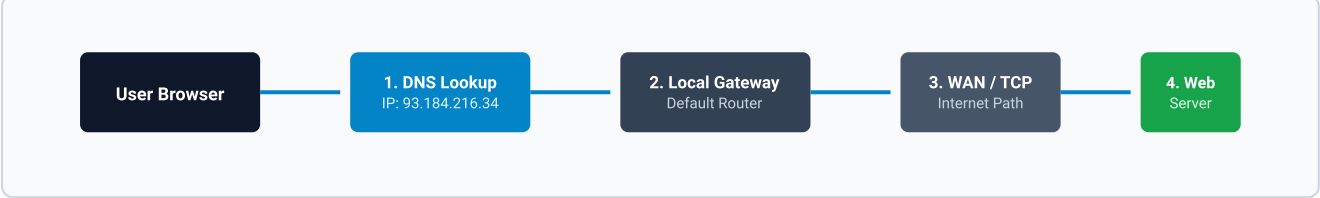
> **Key Difference (Switch vs Router):** A *switch* operates at **Layer 2 (Data Link)** to manage traffic within the *same local network* using MAC addresses. A *router* operates at **Layer 3 (Network)** to forward packets *between separate subnets/networks* using IP addresses.

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Q6. How the Internet Works (10 Marks)

When a user types `example.com` into a web browser, the system executes the following operational steps:

1. **DNS Resolution:** Browser checks local cache. If missing, it queries a Recursive DNS Server, which traverses Root, TLD (.com), and Authoritative servers to resolve `example.com` into IP `93.184.216.34`.
2. **Encapsulation & Handshake:** The OS packages an HTTP/HTTPS request inside TCP segments and IP packets. A 3-Way Handshake (**SYN** → **SYN-ACK** → **ACK**) establishes a reliable connection with the web server.
3. **Routing:** Packets are dispatched to the local Default Gateway/Router, which transmits data across Internet backbones using optimal path routing rules.
4. **Server Response & Rendering:** The destination Web Server processes the HTTP GET request, returns an HTTP `200 OK` response along with web page files, and the browser renders the website.



SECTION C — IP ADDRESSING 20 MARKS

Q7. IPv4 & IPv6 (5 Marks)

IPv6 was introduced primarily to resolve **IPv4 address exhaustion** caused by the global proliferation of Internet devices, while simplifying router processing and header structure.

Feature	IPv4 Address	IPv6 Address
Address Size	32 bits (4 bytes) ≈ 4.3 Billion Addresses	128 bits (16 bytes) ≈ 3.4×10^{38} Addresses
Notation	Dotted Decimal (e.g., `192.168.1.1`)	Hexadecimal (e.g., `2001:0db8:85a3::8a2e:0370:7334`)

Q8. IP Classification (5 Marks)

Target IP Address	Class	Type (Private vs Public)	Scope / Rationale
`10.10.10.5`	**Class A**	**Private**	Falls within standard Private block `10.0.0.0/8`
`172.16.5.20`	**Class B**	**Private**	Falls within standard Private block `172.16.0.0/12`
`192.168.1.50`	**Class C**	**Private**	Falls within standard Private block `192.168.0.0/16`
`8.8.8.8`	**Class A**	**Public**	Globally routable IP (Google Public DNS)

Q9. IP Components Calculation (10 Marks)

Given Input: `192.168.20.45/24`

- **Subnet Mask:** `255.255.255.0` (/24 = 24 ones)
- **Network Address:** `192.168.20.0`
- **Host Portion:** Last Octet = `45`
- **Broadcast Address:** `192.168.20.255`
- **Usable Host Range:** `192.168.20.1` – `192.168.20.254`
- **Total Usable Hosts:** 254 Hosts

Working: /24 prefix defines 24 network bits and 8 host bits ($2^8 - 2 = 254$ usable IPs).

SECTION D — SUBNETTING & CALCULATIONS 25 MARKS

Q10. Basic Subnet Calculation (10 Marks)

Target Subnet: `192.168.10.0/26`

- **Subnet Mask:** `255.255.255.192` (Binary: 11111111.11111111.11111111.11000000)
- **Subnets created from /24:** $2^{(26-24)} = 2^2 = 4$ **Subnets**
- **Usable Hosts per Subnet:** $2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62$ **Hosts**
- **Network Address (Subnet 1):** `192.168.10.0`
- **First Usable Host:** `192.168.10.1` | **Last Usable Host:** `192.168.10.62`
- **Broadcast Address:** `192.168.10.63`

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Q11. Subnetting Design - VLSM Scheme (15 Marks)

Base IP Block: `192.168.10.0/24` | Allocating subnets in descending order of size using Variable Length Subnet Masking (VLSM):

Department	Hosts Req.	Network / CIDR	Subnet Mask	Usable Host Range	Broadcast Address	Max Capa
IT	50	`192.168.10.0/26`	`255.255.255.192`	`192.168.10.1` – `192.168.10.62`	`192.168.10.63`	62 Host
HR	30	`192.168.10.64/27`	`255.255.255.224`	`192.168.10.65` – `192.168.10.94`	`192.168.10.95`	30 Host
Finance	20	`192.168.10.96/27`	`255.255.255.224`	`192.168.10.97` – `192.168.10.126`	`192.168.10.127`	30 Host
Management	10	`192.168.10.128/28`	`255.255.255.240`	`192.168.10.129` – `192.168.10.142`	`192.168.10.143`	14 Host

SECTION E — SCENARIO-BASED QUESTIONS 20 MARKS

Q12. Scenario — Company Network Design (10 Marks)

Design Context: 45 Employees + 5 Servers + 10 Network Devices (60 Total Nodes) → **Local Area Network (LAN)**.

- **Required Devices:** 2x 48-port Managed Switches, 1x Hardware Firewall, 1x Edge Router, Wireless Access Points.
- **Device Roles:**
  - *Switch:* Handles high-speed local data frame switching using internal MAC tables.
  - *Firewall:* Enforces stateful security inspection and blocks unauthorized inbound Internet traffic.
  - *Router:* Connects LAN to WAN, performs Network Address Translation (NAT) for private IPs, and routes traffic.
- **Path to External Website:** Client → Local Switch → Firewall Rule Check → Router (Translates Private IP to Public WAN IP via NAT) → ISP Backbone → Target Web Server.

Q13. Scenario — IP Troubleshooting Checklist (10 Marks)

Symptom: IP `192.168.1.25/24` can access local LAN computers but CANNOT access external websites.
Step-by-step Diagnostic Action Plan:

1. **Verify Local Default Gateway Configuration:** Run `ipconfig` to check if a valid Default Gateway address (e.g., `192.168.1.1`) is set on the NIC.
2. **Test Gateway Reachability:** Ping Default Gateway (`ping 192.168.1.1`). If pings fail, check router connectivity.
3. **Isolate DNS vs Routing:** Ping public IP (`ping 8.8.8.8`) vs domain (`ping google.com`). If IP ping works but domain fails, fix DNS configuration.
4. **Trace External Path:** Execute `tracert 8.8.8.8` to identify exact hop failure point.
5. **Inspect Router WAN & NAT Table:** Access Edge Router admin interface to confirm active WAN connection and NAT translation status.

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